

MA1521 L1 Week 02 Mon 20260817

Evaluate $\int \frac{x}{\sqrt{8-2x^2}} dx$.

Evaluate $\int \sec^8 x \tan x \, dx$.

Evaluate $\int_0^1 (3x + 2)^n dx$ in terms of n .

Evaluate $\int_0^x \frac{dt}{\sqrt{t^2 + 1}}$ as a function of $x > 0$.

Evaluate $\int_0^{\infty} x^3 e^{-x} dx$.

Give the answer as a **fraction** in lowest terms.

Evaluate $\int_0^1 x^2 (1-x)^5 dx$.

Give the answer as a **fraction** in lowest terms.

Evaluate $\int_5^7 \frac{1}{(x-4)(x-2)} dx$.

Give the **exact** answer using logarithms and fractions.

Evaluate $\int_1^{\sqrt{3}} \frac{dx}{x^2 + 1}$.

If $F(x) = \int_x^{x^2} \tan t \, dt$, what is $F'(x)$?

Evaluate $\int_{-1}^1 \frac{\tan^{-1} x}{x^2 + 1} dx$.

Evaluate $\int \frac{1}{(x - 2)(x + 3)} dx$.

Evaluate $\int_1^2 \frac{x^2 + x + 1}{x(x + 8)} dx$.

Evaluate $\int_0^{\pi/2} \frac{1}{(1 + \sin t)^2} dt$.

Evaluate $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$.

Evaluate $\int_e^{\infty} \frac{1}{x (\ln x)^2} dx$.

Evaluate $\int \frac{e^{\tan x}}{\cos^2 x} dx$.

Give the most general answer as an **expression** in terms of x .